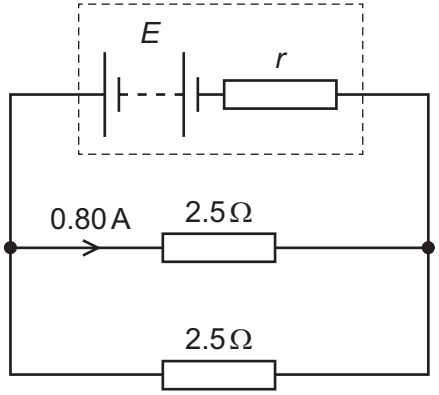
37 A circuit contains a battery of electromotive force E and internal resistance r connected to two resistors each of resistance 2.5 Ω.

The resistors are connected in parallel as shown.



The current in one of the resistors is 0.80 A.

Which expression, where E is in volts and r is in ohms, gives the internal resistance r?

A $\frac{E - 4.0}{1.6}$ **B** $\frac{E - 4.0}{0.80}$ **C** $\frac{E - 2.0}{1.6}$ **D** $\frac{E - 2.0}{0.80}$